

Known Limitations — ITDX26

Honest maturity boundaries; read before evaluation

What this demonstration is not

- Not a hardware demo. No droid enters the room. ITDX26 is an algorithm laboratory; the field footage establishes dataset origin only.
- Not a foundation-model pretraining run. We fine-tune 120 M / 500 M / 1.5 B tiers. We do not pretrain a 2B–7B model in the 8–24 h laptop window.
- Not a biological-prediction claim. FCI (fungal channel interface) is one biosphere signal source contributing to a joint state estimate. Mycelium does not predict.
- Not a proof-of-work claim. DIRTNet is Bitcoin-derived at the structural level (signed observations, pending queues, epoch Merkle roots) but uses no proof-of-work. Permissioned quorum with optional public timestamp anchor.
- Not an acoustic-to-radio translation claim. A hydrophone is not a modem. Translation is claimed only after a dedicated modem/transducer test passes.
- Not a CMMC certification claim. Mycosoft is pursuing CMMC Level 2 (self-assessment in progress, C3PAO audit path underway).
- Not a procurement of the current droid configuration. Army evaluation is for algorithms; procurement is a separate acquisition track.

Known algorithm limitations

- SSM/Mamba training instability at 1.5 B tier — 500 M is ITDX-preferred; 1.5 B runs as stretch demo only.
- Typed relational GNN can over-smooth on very small graphs — mitigated with residual connections and hypergraph incidence.
- MYCA multi-agent latency is bounded at 240 s; abstention is a valid output.
- NLM-Bench splits are grouped by dataset family to prevent leakage; single-run field data is a case study, not a training corpus.
- Fallback ladder degrades gracefully; every fallback state is UI-labeled so a degraded run is never presented as a healthy one.

What we ask

Scoped follow-on with Army-defined unclassified scenario and rubric; SME adjudication of pattern / link / COA products; permission to integrate a field-derived edge dataset into a follow-on experiment; and a funded prototype phase for the AI Analyst / ENVINT workflow.

Morgan Rockcoons
Founder & CEO, Mycosoft, LLC
morgan@mycosoft.org · (619) 483-1077